

The Sharp Scaled Limsup Law for Cosine Families

Full literature-implied answer to arXiv:1310.6202

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Status: literature-implied answer, full scope, likely valid. The scaled cosine question has a sharp affirmative answer. For $a > 0$ the optimal constant is $8/(3\sqrt{3})$; for $a = 0$ it is 2. The semigroup constant is 1. The scaled conclusion follows by combining two theorems of Esterle with a recurrence observation; the supporting authors did not state this limsup implication and a later paper still listed it as open.

1 Source question

Schwenninger and Zwart prove that a strongly continuous cosine family $(C(t))_{t \geq 0}$ on a Banach space satisfies

$$\sup_{t \geq 0} \|C(t) - \cos(at)I\| < 1 \implies C(t) = \cos(at)I.$$

Their final Remark 2.6 asks whether the supremum can be replaced by

$$\limsup_{t \rightarrow \infty} \|C(t) - \cos(at)I\| < 1,$$

or even by a bound with a constant larger than 1. It also says that the semigroup analogue was unclear [1].

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we conclude that

$$\|((q-i)^2 - A)R(q^2 + 1, A)((q+i)^2 - A)\| \leq \frac{q^2}{g(q)} \cdot \frac{2}{1-\delta}.$$

As $q \rightarrow 0^+$, the right-hand-side goes to 0, whereas the left hand side tends to $(I+A)[A-I]^{-1}(I+A)$ as $1 \in \rho(A)$. Therefore, $A = -I$. $\square$

Remark 2.6. We would like to address the open question whether condition (1.4) can be replaced by

$$\limsup_{t \rightarrow \infty} \|C(t) - \cos(at)I\| = r,$$

for some $r < 1$ (or even some constant larger than 1). Even for the classical semigroup case this seems not so clear.

REFERENCES

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- [2] W. Arendt, C. J. K. Batty, M. Hieber, and F. Neubrander. *Vector-valued Laplace transforms and Cauchy problems*, Vol. 96, Monographs in Mathematics. Birkhäuser Verlag, Basel, 2001.
- [3] A. Bobrowski and W. Chojnacki. *Isolated points of the set of bounded cosine families, bounded semigroups. and bounded groups on a Banach space*. Studia Mathematica. 217(3). 219– 241.

Figure 1: Remark 2.6 in arXiv:1310.6202v3, PDF page 6.

The answer below is stronger and sharp. The relation to the supporting literature is an implication identified here: arXiv:1504.02355 explicitly settles the unscaled and semigroup cases, while arXiv:1502.00150 supplies the two structural theorems that settle the remaining scaled case.

2 Supporting results

We use two results from Esterle [2].

1. Theorem 2.3: if $(c_n)_{n \in \mathbb{Z}}$ is a bounded cosine sequence in a Banach algebra and $\text{spec}(c_1)$ is a singleton, then the sequence is scalar: $c_n = \cos(nb)1$ for some $b \in \mathbb{R}$.
2. Lemma 3.5(ii): for $a \geq 0$, if

$$\sup_{t \in \mathbb{R}} |\cos(at) - \cos(bt)| < \frac{8}{3\sqrt{3}},$$

then $b = a$. The constant is optimal for $a > 0$, since

$$\sup_t |\cos(at) - \cos(3at)| = \frac{8}{3\sqrt{3}}.$$

$$(x - y) \operatorname{sinc}\left(\frac{y - x}{2}\right) = 2 \sin\left(\frac{y - x}{2}\right) = 0.$$

Since $\chi(\operatorname{sinc}(\frac{y-x}{2})) = \operatorname{sinc}(0) = 1$ for every $\chi \in \hat{A}$, $\operatorname{sinc}(\frac{y-x}{2})$ is invertible and $x = y$.

If $\lambda \in \pi\mathbb{Z}$, then we have $\cos(x) = \cos(y)$, which gives

$$(y^2 - x^2) \operatorname{sinc}\left(\frac{y - x}{2}\right) \operatorname{sinc}\left(\frac{x + y}{2}\right) = 4 \sin\left(\frac{y - x}{2}\right) \sin\left(\frac{x + y}{2}\right) = 0.$$

We see again that $\operatorname{sinc}(\frac{y-x}{2})$ and $\operatorname{sinc}(\frac{x+y}{2})$ are invertible, which shows that $x^2 = y^2$.

Theorem 2.3. *Let $(c_n)_{n \in \mathbb{Z}}$ be a bounded cosine sequence in a Banach algebra A , and assume that $\text{spec}(c_1)$ is a singleton. Then $(c_n)_{n \geq 1}$ is a scalar sequence, and there exists $b \in \mathbb{R}$ such that $c_n = \cos(nb) \cdot 1_A$ for $n \geq 1$.*

Proof: Set $M = \sup_{n \geq 1} \|c_n\|$. We can assume that A is a commutative Banach algebra generated by c_1 , so that $\hat{A}$ consists of a single character χ . Let $\lambda_n = \chi(c_n)$ be the unique element of $\text{spec}(c_n)$, and set $\beta = \beta_\chi = \arccos(\chi(c_1)) = \chi(\arccos(c_1))$, so that $\lambda_n = \cos(n\beta)$ for $n \geq 1$.

Set $x_n = \arccos(c_n)$. Since $c_n = \cos(x_n)$, we have $\cos(n\beta) = \cos(\chi(x_n))$. It follows then from standard properties of the cosine function on $\mathbb{C}$ that there exists

Figure 2: Esterle's singleton-spectrum rigidity theorem, arXiv:1502.00150v2, PDF page 6.

$$= \left| \cos\left(\left(an + \frac{a}{2k}\right)p\right) + \cos\left(\left(an + \frac{a}{2k}\right)q\right) \right| = 2.$$

Hence $k = 0$ if $\sup_{t \in \mathbb{R}} |\cos(ta) - \cos(tb)| < 2$. $\square$

Lemma 3.5. *Let $a \geq 0$, and set $\Omega(a, m) := \{b \geq 0 \mid \sup_{t \in \mathbb{R}} |\cos(ta) - \cos(tb)| \leq m\}$ for $m \geq 0$.*

(i) If $m < 2$, then $\Omega(a, m)$ is finite, and every $b \in \Omega(a, m)$ has the form $b = \frac{pa}{q}$, where p, q are odd, $\gcd(p, q) = 1$, $1 \leq p \leq \frac{\pi}{\arccos(m-1)}$, $1 \leq q \leq \frac{\pi}{\arccos(m-1)}$.

(ii) If $m < \frac{8}{3\sqrt{3}}$, then $\Omega(a, m) = \{a\}$.

Proof: Assume that $m < 2$. Since $\Omega(0, m) = \{0\}$, we can assume that $a > 0$. Let $b \in \Omega(a, m)$. Then $b \neq 0$. We may restrict attention to the case where $a = 1$, and there exists positive integers p and q such that $b = \frac{p}{q}$, with $\gcd(p, q) = 1$. If p or q were even, we would have $\sup_{t \in \mathbb{R}} |\cos(t) - \cos(bt)| \geq \sup_{n \geq 1} |\cos(np\pi) - \cos(nq\pi)| = 2$, and so p and q are odd.

It follows from Bezout's theorem that there exists $(u, v) \in \mathbb{Z}$ such that $2up - (2v + 1)q = 1$. We have

$$\begin{aligned} \sup_{t \in \mathbb{R}} |\cos(t) - \cos(bt)| &\geq \cos(2u\pi) - \cos\left(\frac{2up\pi}{q}\right) = 1 - \cos\left((2v + 1)\pi + \frac{\pi}{q}\right) \\ &= 1 + \cos\left(\frac{\pi}{q}\right). \end{aligned}$$

Hence $q \leq \frac{\pi}{\arccos(m-1)}$. Since $\sup_{t \in \mathbb{R}} |\cos(t) - \cos(bt)| = \sup_{t \in \mathbb{R}} |\cos(t) - \cos(\frac{t}{b})|$,

Figure 3: Esterle's sharp scalar separation lemma, arXiv:1502.00150v2, PDF page 9.

3 The sharp answer

Theorem 1 (Sharp scaled limsup law). *Let $(C(t))_{t \in \mathbb{R}}$ be a strongly continuous cosine family on a complex Banach space X .*

1. *If $a > 0$ and*

$$L := \limsup_{t \rightarrow \infty} \|C(t) - \cos(at)I\| < \frac{8}{3\sqrt{3}},$$

then $C(t) = \cos(at)I$ for every $t \in \mathbb{R}$. The constant $8/(3\sqrt{3})$ is optimal.

2. *If $a = 0$ and*

$$\limsup_{t \rightarrow \infty} \|C(t) - I\| < 2,$$

then $C(t) = I$ for every $t \in \mathbb{R}$, and 2 is optimal.

Proof. The second assertion is exactly Theorem 2.5 of Schwenninger–Zwart [3]; optimality follows from any nonconstant scalar cosine.

We prove the first assertion. The tail hypothesis bounds $C(t)$ for all sufficiently large t . Strong continuity and the uniform boundedness principle bound it on each compact interval, while $C(-t) = C(t)$. Hence

$$M := \sup_{t \in \mathbb{R}} \|C(t)\| < \infty.$$

Let A be the closed unital subalgebra of $\mathcal{B}(X)$ generated by the family. The cosine identity makes A commutative. Fix a character χ of A . Then

$$c_\chi(t) := \chi(C(t))$$

is a bounded scalar cosine function.

We first note a standard scalar dichotomy. If a bounded scalar cosine function c on $\mathbb{R}$ is discontinuous, then for every continuous scalar cosine $t \mapsto \cos(at)$,

$$\limsup_{t \rightarrow \infty} |c(t) - \cos(at)| = 2. \tag{1}$$

One way to see this is to write $c(t) = (\gamma(t) + \gamma(t)^{-1})/2$ for a group character $\gamma : \mathbb{R} \rightarrow \mathbb{T}$. If the closure of $t \mapsto (e^{iat}, \gamma(t))$ is a proper subgroup of $\mathbb{T}^2$, an integer relation makes a nonzero power of γ continuous; divisibility of $\mathbb{R}$ then makes γ continuous. Otherwise the image is dense in $\mathbb{T}^2$. The same is true on every positive tail, by recurrence in the compact group, and (1) follows. For $a = 0$ the identical argument uses only the second coordinate. This is also the infinity-tail counterpart of Esterle’s discontinuous-scalar density proposition [2, Proposition 3.1].

Since

$$\limsup_{t \rightarrow \infty} |c_\chi(t) - \cos(at)| \leq L < 2,$$

the dichotomy shows that c_χ is continuous. Thus $c_\chi(t) = \cos(b_\chi t)$ for some $b_\chi \geq 0$. A finite sum of continuous scalar characters is recurrent, so

$$\sup_{t \in \mathbb{R}} |\cos(b_\chi t) - \cos(at)| = \limsup_{t \rightarrow \infty} |\cos(b_\chi t) - \cos(at)| \leq L.$$

Esterle’s Lemma 3.5(ii) now gives $b_\chi = a$. Therefore, for every $s \in \mathbb{R}$ and every character χ ,

$$\chi(C(s)) = \cos(as),$$

and hence

$$\text{spec}_A(C(s)) = \{\cos(as)\}. \quad (2)$$

Fix s . The sampled family $c_n = C(ns)$ is a bounded cosine sequence in A , and its first element has singleton spectrum by (2). Esterle's Theorem 2.3 says that this sequence is scalar. In particular $C(s)$ is scalar, and its unique spectral value in (2) forces

$$C(s) = \cos(as)I.$$

Since s was arbitrary, the desired identity holds on all of $\mathbb{R}$.

Finally, when $a > 0$, the scalar cosine family $C(t) = \cos(3at)I$ is nontrivial and has limsup distance exactly $8/(3\sqrt{3})$. Thus no larger strict-threshold constant is possible. $\square$

Theorem 2 (Sharp semigroup limsup law). *Let $(T(t))_{t \geq 0}$ be a semigroup in a unital normed algebra. Then*

$$\limsup_{t \rightarrow \infty} \|T(t) - I\| < 1 \implies T(t) = I \quad (t \geq 0).$$

The constant 1 is optimal.

Proof. This is Theorem 3.1 and Remark 3.2 of Schwenninger–Zwart [3]. Sharpness is witnessed by the scalar semigroup $T(t) = e^{-t}$, for which the limsup distance is 1. $\square$

algebra. If $\|C(2s) - I\| \leq 2$ and that $\rho(C(s) - I) < 1$ for some $s \in \mathbb{R}$, Then,

$$C(s) = \sqrt{I - \frac{I - C(2s)}{2}},$$

where the square root is defined as described above.

With the above preparatory results, the [limsup- \$\infty\$ -law](#) is now easy to show. The proof can be done analogously to the one in [\[6\]](#) Theorem 3.2], which in turn can be seen as an elegant refinement of the technique used in the *three-lines-proof* in [\[1\]](#).

Theorem 2.5. *Let $(C(t))_{t \in \mathbb{R}}$ be a cosine family in a unital Banach algebra A . Then, $\limsup_{t \rightarrow \infty} \|C(t) - I\| < 2$ implies that $C(t) = I$ for all $t \in \mathbb{R}$.*

Proof. By Proposition [2.3](#) we have that $\rho(C(t) - I) = 0$ for $t \in \mathbb{R}$. Thus, we can apply Lemma [2.4](#) and Eq. [\(2.5\)](#) so that for all $s \in \mathbb{R}$,

$$\|I - C(s)\| \leq 1 - \sqrt{1 - \left\| \frac{I - C(2s)}{2} \right\|} \leq 1.$$

With $S := \limsup_{s \rightarrow \infty} \|C(s) - I\|$, this yields that

$$S \leq 1 - \sqrt{1 - \frac{S}{2}} \leq 1,$$

which implies that $S = 0$. Hence, Lemma [2.1](#) concludes the assertion. $\square$

Remark 2.6. After finishing this note, Esterle pointed out that, alternatively, [\[5\]](#) Theorem 2.3] implies that for a bounded cosine sequence with $\rho(C(1) - I) = 0$, it follows that $C(t) = \cos(at)I$ for all $t \in \mathbb{R}$ and some $a \in \mathbb{R}$. Thus, $\limsup_{t \rightarrow \infty} \|C(t) - I\| < 2$ implies $C(t) = I$ for all $t \in \mathbb{R}$ and therefore the proof of Theorem [2.5](#) can be

Figure 4: The sharp unscaled cosine result, arXiv:1504.02355v1, PDF page 3.

The proof is based on an elegant idea of Arendt, which can directly be applied to weaken the sup to limsup in the theorem.

Theorem 2.9. *Let $(C(n))_{n \in \mathbb{Z}}$ be a discrete cosine family in a unital Banach algebra. Then,*

$$\limsup_{n \rightarrow \infty} \|C(n) - I\| < \frac{3}{2} \implies C(n) = I \quad \forall n \in \mathbb{Z}.$$

This result is optimal as can be seen by $C(n) = \cos(\frac{2n\pi}{3})$ which yields $\limsup_{n \rightarrow \infty} \|C(n) - I\| = \frac{3}{2}$, see [9 Theorem 3.2].

3. THE CORRESPONDING SEMIGROUP RESULT

Let us finally state the corresponding result for (discrete) semigroups in a unital normed algebra, which is a corollary of a well-known result by Wallen [10].

Theorem 3.1. *Let $\{T_n\}_{n \in \mathbb{N}}$ be a semigroup in a normed unital algebra. Then,*

$$(3.1) \quad \limsup_{n \rightarrow \infty} \|T_n - I\| < 1 \implies T_n = I \quad \forall n \in \mathbb{N}.$$

Proof. If $\limsup_{n \rightarrow \infty} \|T_n - I\| < 1$, then $\liminf_{n \in \mathbb{N}} \frac{1}{n} \sum_{j=1}^n \|T_j - I\| < 1$. By Wallen [10], the assertion follows. $\square$

Remark 3.2. Clearly, Theorem 3.1 implies that for a semigroup on $[0, \infty)$, $(T(t))_{t \geq 0}$, we have that

$$(3.2) \quad \limsup \|T(t) - I\| < 1 \implies T(t) = I \quad \forall t \geq 0.$$

Figure 5: The semigroup limsup result, arXiv:1504.02355v1, PDF page 4.

4 Status, scope, and search bounds

This settles every case mentioned in Remark 2.6, and it gives the best possible constants. The provenance is *literature-implied*, not an explicit later answer: Schwenninger–Zwart [3] explicitly settle $a = 0$ and semigroups, but still describe the scaled law as a related open question. Esterle [2] proves the two decisive ingredients, but does not state their tail-limsup consequence. The short character-spectrum bridge in the proof above identifies that consequence.

A bounded search used the exact formula, the source id and title, author names, and the phrases “scaled cosine family”, “limsup to infinity”, and “scalar cosine family”. It found arXiv:1504.02355, arXiv:1502.00150, and arXiv:1505.06064 (the discrete-group analogue), but no later paper explicitly stating the continuous scaled limsup theorem. Novelty is therefore claimed only for the identification, not for the two supporting structural theorems.

Human review should focus on the scalar discontinuity dichotomy (1) and on applying Esterle’s Theorem 2.3 inside the generated commutative Banach algebra. Both remaining steps are direct.

References

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